Example Homework Write-Ups

Combinatorics

MATH 31003-01 - Fall 2026

This page shows what a complete handwritten homework solution should look like. The goal is not a long proof. For each exercise, write at least one sentence that explains your approach, and give word labels to the quantities in your calculation.

The model solutions are set in larger, handwriting-like type to suggest the amount of work that you could reasonably write by hand.

0.1 Example assignment

0.1.1 Exercise 1: Forming a student council

Points: 10

A student organization has 10 mathematics majors and 8 computer science majors. It will choose one president, one secretary, and three ordinary representatives. No student may hold two positions. The president and secretary must have different majors. The three ordinary-representative positions are not distinguished, and those representatives may have either major. How many councils are possible?

0.1.2 Exercise 2: Seven-digit numbers with repetition

Points: 10

How many seven-digit positive integers have at least one repeated digit? Recall that a seven-digit integer cannot begin with 0.

0.2 Full-credit solutions

These solutions include the required words but are intentionally concise.

0.2.1 Exercise 1

Approach. *Choose the two officers first and then choose the three ordinary representatives from the students who remain.*

Phase 1 --- officers. *If the president is a mathematics major, there are* $10 \cdot 8$ *choices; reversing the majors gives* $8 \cdot 10$ *choices. Thus the officers can be chosen in*

$$10 \cdot 8 + 8 \cdot 10 = 160$$

ways.

Phase 2 --- ordinary representatives. *After choosing the officers, 16 students remain, so the three unordered representatives can be chosen in* $C(16, 3) = 560$ *ways.*

Every council is obtained once by these two phases, so the product rule gives

$$160 \cdot C(16, 3) = 160 \cdot 560 = 89{,}600.$$

0.2.2 Exercise 2

Approach. *Count all seven-digit integers and subtract the ones whose digits are all different.*
Universe U --- all seven-digit integers. *The first digit has 9 choices and each other digit has 10 choices, so*

$$|U| = 9 \cdot 10^6 = 9{,}000{,}000.$$

Forbidden F --- no digit repeats. *The first digit has 9 choices. The remaining digits then have $9, 8, 7, 6, 5, 4$ choices in order, so*

$$|F| = 9 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 544{,}320.$$

Therefore the difference rule gives

$$|U| - |F| = 9{,}000{,}000 - 544{,}320 = 8{,}455{,}680.$$

0.3 Correct but incomplete solutions

The calculations and final answers below are correct, but these solutions would not earn full credit.

0.3.1 Exercise 1

$$(10 \cdot 8 + 8 \cdot 10) \cdot C(16, 3) = 160 \cdot 560 = 89{,}600.$$

This is incomplete because it does not say what is chosen in each phase, what the two officer terms count, or why the two main quantities are multiplied.

0.3.2 Exercise 2

Approach. *Count all seven-digit integers and subtract the ones whose digits are all different.*
Universe U --- all seven-digit integers. *There are $9 \cdot 10^6 = 9{,}000{,}000$ such integers.*
Forbidden F --- no digit repeats. *The first digit has 9 choices. The remaining digits then have $9, 8, 7, 6, 5, 4$ choices in order, so*

$$|F| = 9 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 544{,}320.$$

Therefore

$$|U| - |F| = 9{,}000{,}000 - 544{,}320 = 8{,}455{,}680.$$

This solution has a useful approach sentence and identifies both sets. It is still incomplete because it merely states $|U| = 9 \cdot 10^6$ without explaining why the first factor is 9 or what the exponent 6 represents.

The first incomplete example gives only a formula. The second is much closer to complete, but it leaves one important part of its formula unexplained. Both show that the words must account for the whole model, not merely appear somewhere in the solution.
The labels do not have to match these examples exactly. They may be written beside a formula or on separate lines, and abbreviations are fine after they are defined. What matters is that a reader can tell what each important number counts and why the displayed operations match the approach.

0.4 Sample Rubric

Each exercise is graded out of 10 points.

- **2 points: approach and modeling.** At least one sentence explains what is being counted and why the chosen rule applies.
- **2 points: intermediate quantities and factors.** One point is for giving the important quantities word labels and meanings, such as **Phase 1 — officers** and **Phase 2 — representatives**, or **Universe — all seven-digit integers** and **Forbidden — no digit repeats**. One point is for explaining any non-obvious terms or factors in those quantities.
- **4 points: intermediate counts.** The two main quantities are counted correctly, worth 2 points each.
- **1 point: combination.** The quantities are combined correctly using the product, sum, or difference rule.
- **1 point: final answer.** The numerical answer is clearly stated.

The full-credit models receive 10 out of 10. The incomplete solution to Exercise 1 receives 6 out of 10: it earns the calculation, combination, and final-answer points, but none of the communication points. The incomplete solution to Exercise 2 receives 9 out of 10: its approach and labels are clear, but it loses one point because the factors counting U are not explained.
A correct bare formula is therefore not a full-credit solution. On the other hand, no long explanation is required: one useful approach sentence plus short word labels for the main quantities is enough. Equivalent labels and wording are welcome; students do not need to copy the exact format above.